

Daniel Seungmin Lee

Undergraduate Researcher

Daniel Seungmin Lee
University of California, Berkeley, CA 94720, USA

Citizenship: United States of America, Republic of Korea
Languages: English, Korean

(+82) 010-3712-0224
dlee2026@berkeley.edu

Skills

- Experimental/Theoretical Physics Research
- Mathematical Modeling
- Mathematical Maturity
- Coding
- Problem Solving
- Leadership

I have experience in both experimental and theoretical condensed matter physics research, specifically on correlated and topological phenomena in synthetic materials. My mathematical interests are in topology and their appearance in modern condensed matter theory.

Education

- **Double Major in Physics and Mathematics**
University of California, Berkeley
GPA: 3.92/4.0

University of California, Berkeley / 3rd year undergraduate
August, 2022 - Present

I'm currently declared as a double major in math and physics in the College of Letters & Sciences. I'll be returning to Berkeley this Fall as a third year student, planning to graduate this academic cycle.

Experience

Crommie Group

August, 2023 - August, 2024

I contributed to the group's work on Scanning Tunneling Spectroscopy measurements of twisted 2D materials – transition metal dichalcogenides, graphene, etc. I mainly participated in the device fabrication process.

Related Skills/Experience

- Exfoliating Nanomaterials
- Polymer Film Fabrication
- Stacking
- Glovebox
- Gating 2D Samples
- Lithography
- Polymer Annealing
- Oxygen Plasma

Military Service (Republic of Korea Air Force)

September, 2024 - Present

I'm currently serving as a Chemical, Biological, Radiological, Nuclear (CBRN) specialist in the Republic of Korea Air Force. I will be discharged in June, 2026.

Joint Theoretical Research with Dr. Junyeong Ahn

November, 2025 - Present

I have conducted independent theoretical studies in condensed matter field theory. From June, 2025, I've narrowed my focus to topological condensed matter and superconductivity, landing on the topic of topological multiterminal Josephson junctions. I have since completed a paper titled "Real-Bundle Topology in Multiterminal Josephson Junctions"

Related Skills/Experience

- Numerical Topology
- Python, Matlab
- Theoretical Modeling

Quantum Devices Group @ UC Berkeley

June, 2026 - Present

I'm currently working on a project on "junction readout" of superconducting qubits, where the shunting capacitance between a readout resonator and a transmon in dispersive readout is replaced by a coupling Josephson junction. It is predicted that this readout mechanism highly suppresses purcell decay, achieving high T_1 .

Related Skills/Experience

- Superconducting chip design
- Numerical methods
- Superconducting qubit theory

Honors

Summer Undergraduate Research Fellowship (SURF)

June 2024 - August 2024, University of California, Berkeley Letters & Sciences

I was selected as a SURF fellow under the Math & Physical Sciences category for my project "Detecting Correlated States In Magic-Angle Helical Trilayer Graphene." With the stipend, I was able to conduct independent experimental condensed matter research under the guidance of Dr. Michael Crommie.